

GATES: Group-Aware Time-Efficient Stopping for Longitudinal Biological Experiments

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Abstract

Longitudinal biological experiments are commonly observed on fixed schedules even when their endpoints become predictable earlier. Prediction alone, however, does not establish that continued measurement is unnecessary: confidence can fail under experiment-to-experiment heterogeneity. We introduce GATES, an experiment-disjoint selective-stopping framework that asks whether the available prefix of a trajectory justifies an early endpoint decision. GATES combines prefix-based prediction, calibration on independent biological experiments, a feature-distance refusal check, and sequential STOP, CONTINUE, or ABSTAIN actions. We evaluate the complete frozen policy retrospectively on 988 medaka retinal organoids from 11 independent experiments, comprising 114,510 released longitudinal morphometric observations through 72 h. At the frozen operating point, GATES made 439 early decisions with 25 errors: a 5.69% observed held-out erroneous-early-stop rate (exact unit-level 95% confidence interval, 3.72%–8.29%), 44.43% coverage, and 12.26% retrospectively estimated observation savings. Removing refusal increased observed error to 8.60% while increasing estimated savings to 13.65%. The aggregate result conceals substantial heterogeneity: one held-out experiment reached 30.61% error among early stops, and experiment-bootstrap uncertainty is wide. These findings are promising retrospective evidence, not an anytime-valid or deployment guarantee. GATES provides an auditable formulation for converting longitudinal prediction into a decision about whether another measurement remains necessary.

1 Introduction

Longitudinal biological studies often follow fixed acquisition schedules: images or measurements are collected repeatedly until a predetermined endpoint, even when the available trajectory may already contain useful endpoint information. Earlier decisions could reduce acquisition burden, microscope occupancy, storage, and downstream analysis for se-

lected experimental units. A prematurely wrong decision, however, can corrupt the study by stopping observation before the true endpoint is known. The relevant problem is therefore not simply early prediction; it is deciding when an early prediction is reliable enough to justify termination.

This distinction is especially important in biology. Measurements are nested within experimental units, which are themselves nested within batches or experiments. A classifier can perform well in aggregate while failing severely in one held-out experiment. Confidence estimated without respecting this hierarchy may therefore be unsuitable for an intervention such as stopping. Experimental groups are not merely train–test bookkeeping: experiment-to-experiment variation is part of the reliability problem.

GATES frames observation as a sequential selective decision. At each candidate time, it uses only the available trajectory prefix, estimates the eventual binary endpoint, checks calibrated confidence and familiarity, and returns STOP, CONTINUE, or ABSTAIN. The policy is evaluated as a whole on experiments excluded from fitting, calibration, and refusal-threshold selection. Its principal outputs are erroneous early-stop rate, coverage, and retrospectively estimated observation savings, reported across complete risk–savings and risk–coverage frontiers rather than as a single accuracy value.

Our contributions are:

- an experiment-disjoint formulation of selective stopping for longitudinal biological endpoints;
- independent-group confidence calibration coupled to explicit trajectory-level refusal;
- evaluation through risk–coverage–savings trade-offs with unit- and experiment-level uncertainty; and
- a reproducible decision record that retains negative transfer, robustness failures, and severe experiment-specific errors.

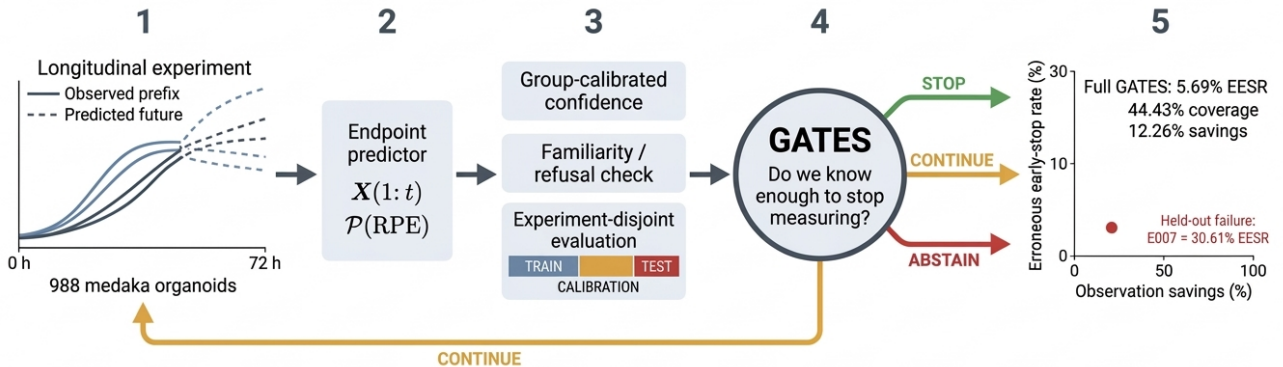


Figure 1. Overview of GATES. Partial longitudinal trajectories are converted into endpoint predictions, calibrated using independent experimental groups, and passed through a refusal mechanism before a sequential STOP, CONTINUE, or ABSTAIN decision. Evaluation is performed on completely held-out experiments and summarized through the tradeoff between premature-decision risk and retrospectively estimated observation savings.

2 Related work

Early classification and selective prediction.

Early classification of time series studies the trade-off between predictive accuracy and decision earliness, including rules for selecting a useful prefix (Xing et al., 2012). Selective prediction and rejection learning permit abstention so that risk is measured on the covered subset (Geifman and El-Yaniv, 2019). GATES draws on both traditions, but treats complete biological experiments as the calibration and generalization boundary and measures the downstream cost of continued observation.

Calibration and risk control. Conformal prediction and conformal risk control provide rigorous uncertainty or risk statements under explicit assumptions (Angelopoulos and Bates, 2023; Angelopoulos et al., 2024). The current GATES policy is not conformal and does not provide distribution-free, anytime-valid, or group-conditional risk control. Its observed held-out rate is an empirical property of the complete frozen sequential procedure.

Experimental stopping. Confidence-bound early stopping (CBES) is the closest prior experimental-stopping method (Liu et al., 2026). CBES combines probabilistic prediction, con-

fidence bounds, sequential calibration, and a Hoeffding-based validation bound in its own setting. We do not claim a stronger formal result. The comparator used here is a documented binary adaptation: bootstrap logistic ensembles replace the original continuous Gaussian-process model, and symmetric bounds around 0.5 define the two endpoint decisions. Both receive the same features, candidate times, and held-out experiments. This is not a byte-for-byte reproduction of all CBES formulations.

Biological heterogeneity. Technical and biological batch effects can invalidate naive generalization (Leek et al., 2010). Reporting standards for supervised learning in biology similarly emphasize separation of development and evaluation data, transparent optimization, and appropriate metrics (Walsh et al., 2021). The source orgAInoid work established early outcome prediction from retinal organoid morphology (Afting et al., 2026); GATES addresses the downstream question of whether a specific partial trajectory justifies terminating observation.

3 Dataset and biological context

We use the CC BY 4.0 source data released by Afting et al. (2026). The biological system is

Table 1. Dataset and evaluation hierarchy. Observations are not independent biological units.

Released morphometric observations	114,510
Organoid wells (prediction units)	988
Independent experiments (holdout units)	11
Maximum observation horizon	72 h
RPE positive / negative	405 / 583
Lens positive / negative	450 / 538
Complete 144-observation trajectories	381
Trajectories with fewer than 72 observations	189
Trajectories with fewer than 12 observations	85

Oryzias latipes (medaka) retinal organoids, not human organoids. The study contains 988 organoid wells from 11 independent experiments, imaged every 30 minutes for up to 72 h. The final morphology label for retinal pigment epithelium (RPE) formation is the primary binary endpoint; Lens formation is retained as a transfer analysis.

The hierarchy is 114,510 longitudinal observations nested within 988 organoid wells, themselves nested within 11 experiments. An organoid well is the prediction unit; an experiment is the generalization and resampling boundary. Availability variables are excluded from predictors because they can encode experiment identity.

3.1 Reconciling image and morphometric counts

The source article reports 117,249 acquired images, whereas GATES uses exactly 114,510 released morphometric rows. These counts describe different objects. The pipeline pins and reads `Extended_Data_2.csv`, the publication’s released morphometrics table. Audit and modeling loaders select columns but remove no rows. The difference of 2,739 therefore arose upstream, between acquisition and the published table. Neither the article nor the table gives per-image exclusion reasons; no specific cause, such as segmentation failure, is inferred.

4 Problem formulation

For organoid i , let $X_{i,1:t}$ denote the measurements available up to candidate time t , and let $y_i \in \{0, 1\}$ be its final endpoint at horizon $T = 72$ h. A predictor produces $p_{i,t} = \Pr(y_i = 1 \mid X_{i,1:t})$ and label

$$\hat{y}_{i,t} = \mathbb{K}[p_{i,t} \geq 1/2]. \quad (1)$$

The sequential policy selects a terminal action $a_{i,t} \in \{\text{STOP}, \text{CONTINUE}, \text{ABSTAIN}\}$ and an associated de-

cision time τ_i . An erroneous early stop occurs if

$$\tau_i < T \quad \wedge \quad \hat{y}_{i,\tau_i} \neq y_i. \quad (2)$$

Let $S_i = \mathbb{K}[\tau_i < T \wedge a_{i,\tau_i} = \text{STOP}]$. The erroneous early-stop rate is

$$\text{EESR} = \frac{\sum_{i=1}^n S_i \mathbb{K}[\hat{y}_{i,\tau_i} \neq y_i]}{\sum_{i=1}^n S_i}. \quad (3)$$

Coverage is the fraction of units stopped early,

$$\text{Coverage} = \frac{1}{n} \sum_{i=1}^n S_i. \quad (4)$$

If $m_i(T)$ is the number of scheduled observations available through the endpoint and $m_i(\tau_i)$ the number consumed by the policy, retrospective observation savings are

$$\text{Savings} = \frac{1}{n} \sum_{i=1}^n S_i \left(1 - \frac{m_i(\tau_i)}{m_i(T)} \right). \quad (5)$$

This quantity estimates observations that would not have been required after a frozen early stop; it is not observed microscope time, laboratory duration, monetary cost, or prospective resource savings.

5 GATES method

Figure 1 summarizes the complete decision path. At each candidate time, leakage-safe prefix features summarize only measurements available by that time. A regularized classifier estimates the endpoint probability. Identifiers, availability variables, and final labels are excluded from predictor features.

5.1 Independent-group calibration

Training, preprocessing, confidence calibration, and refusal-threshold fitting exclude the complete held-out experiment. Development experiments calibrate confidence thresholds for the requested operating point. If confidence does not pass, the action is CONTINUE. The 5% label is a calibration target, not certified risk control.

5.2 Familiarity and refusal

A training-derived feature-distance score compares the current prefix with the fitted reference distribution. If the score exceeds its frozen threshold, the policy returns ABSTAIN and observes the unit through the endpoint. Otherwise, a sufficiently confident endpoint call returns STOP. The refusal score is a limited diagnostic for generic feature-space distance; it does not prove that every shifted experiment will be detected.

5.3 Sequential decision contract

Repeated looks occur at the candidate times of the frozen policy. STOP commits to an early endpoint prediction, CONTINUE requests further observation, and ABSTAIN disables early automation for that trajectory. Because the procedure does not construct an anytime-valid confidence sequence, Eq. (3) remains an empirical held-out metric for the full sequential policy.

6 Experiment-disjoint evaluation

Eleven rotations hold out one complete experiment at a time. Every well, frame, prefix, and label from that experiment is excluded from model fitting and rule selection. All prefixes from a well remain inside the same experiment. The rotations yield exactly one held-out terminal decision for each of 988 organoids.

Unit-level EESR receives a two-sided Clopper–Pearson interval. To reflect variation at the actual generalization boundary, aggregate metrics also receive a 10,000-draw percentile bootstrap over the 11 held-out experiments. The experiment-level interval is necessarily wide and is more relevant to between-experiment transfer than the unit-level binomial interval alone.

7 Baselines

The frozen comparison includes full observation, fixed stopping times, naive confidence, non-group calibration, a CBES-style adaptation, GATES without refusal, and Full GATES. Each method is evaluated on the same held-out experiments. Fixed time applies one decision time to all eligible trajectories. Naive confidence uses uncalibrated predictive confidence. Non-group calibration pools development units without preserving experiments as calibration groups. The CBES-style comparator uses sequential calibration and validation roles as described above. Full GATES adds experiment-aware calibration and trajectory-level refusal.

No single scalar ranking is adequate because the methods stop different numbers of organoids at different times. Risk–savings and risk–coverage frontiers are therefore the primary comparison (figure 2).

8 Main results

At the frozen 5% operating point, Full GATES stopped 439 of 988 organoids and made 25 errors. Observed held-out EESR was 5.69% (unit-level exact 95% confidence interval, 3.72%–8.29%),

coverage was 44.43%, abstention was 3.85%, and retrospective observation savings were 12.26%. Experiment-bootstrap intervals were 0.68%–12.60% for EESR, 27.06%–61.52% for coverage, and 6.74%–17.98% for savings. The contrast between the unit-level interval and the wider experiment bootstrap underscores that the 114,510 observations are not independent evidence for cross-experiment generalization.

At this point, GATES permits early automation for 44.43% of organoids and deliberately continues observation for the remainder rather than forcing a premature decision. At the frozen 10% point, coverage and savings increased to 58.70% and 23.63%, respectively, while observed EESR increased to 13.10%. Coverage is therefore an operating characteristic, not the fraction on which prediction “works.”

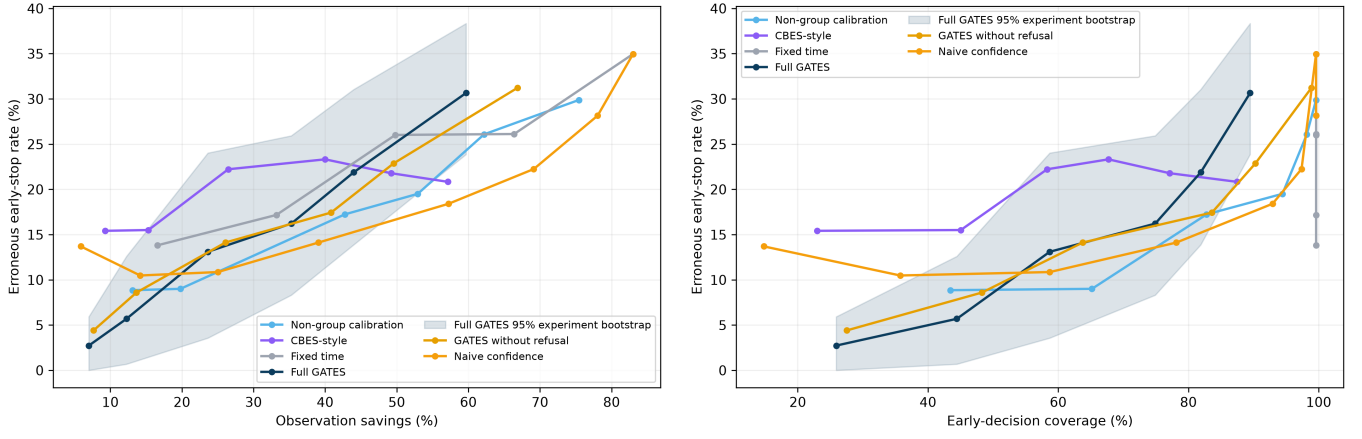
9 Refusal ablation

Without refusal, GATES made 477 early decisions with 41 errors, corresponding to 8.60% EESR and 13.65% savings. With refusal, it made 439 early decisions with 25 errors, corresponding to 5.69% EESR and 12.26% savings (figure 3). Thus, under the frozen RPE evaluation, refusal reduced observed premature error while sacrificing 1.39 percentage points of estimated observation savings. The effect is operational rather than decorative, but it does not establish general OOD detection: the unit-level gate missed the important E007 and E012 failure modes.

10 Experiment-level heterogeneity

Seven held-out experiments had zero early-stop errors; E007, E008, E009, and E012 did not (figure 4). E007 contributed 15 errors among 49 early stops (30.61% EESR), including five false positives and ten false negatives at 24, 48, and 60 h. Generic feature-distance statistics were not extreme, and none of the erroneous stops crossed the unit-level OOD threshold. The supported diagnosis is possible conditional or concept shift; its biological mechanism is undetermined.

E012 produced six false-positive errors at 48 h. Their mean predictive confidence was high and ensemble disagreement was low. At 12 h, mean predicted probability was 0.247 despite a 54.55% positive prevalence, with Brier score 0.362. The primary numerical diagnosis is calibration failure; concurrent conditional shift cannot be excluded. These experiments show why aggregate confidence and aggregate risk cannot establish group-level reliability.



(a) Observed EESR versus retrospectively estimated observation savings.

(b) Observed EESR versus early-decision coverage.

Figure 2. Frozen RPE operating frontiers across 11 experiment-held-out rotations. Marker shape, line style, and color distinguish methods. The emphasized Full GATES point is the frozen 5% operating point. Methods occupy different coverage and savings regimes, so the tradeoff surface, rather than a single bar, is the appropriate comparison.

Table 2. Frozen RPE comparison at the selected 5% operating point. Values describe different risk–coverage–savings tradeoffs and should not be read as a simple accuracy ranking.

Method	Early stops	Errors	EESR	Coverage / savings
Safest tested fixed time	984	136	13.82%	99.60% / 16.60%
Naive confidence	643	67	10.48%	65.08% / 14.17%
Non-group calibration	644	58	9.01%	65.18% / 19.79%
CBES-style adaptation	445	69	15.51%	45.04% / 15.32%
GATES without refusal	477	41	8.60%	48.28% / 13.65%
Full Gates	439	25	5.69%	44.43% / 12.26%

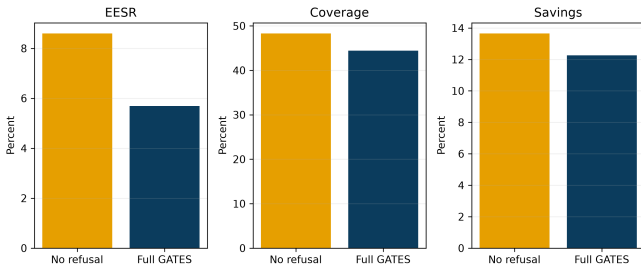


Figure 3. Refusal ablation on frozen held-out RPE decisions. Adding the feature-distance refusal check reduced errors from 41/477 to 25/439 while estimated savings changed from 13.65% to 12.26%. Both risk and retained coverage are reported to avoid rewarding rejection alone.

11 Robustness and negative results

The primary policy did not remain below its target label under several perturbations (table 3). Irregular sampling was particularly damaging, reaching 10.61% EESR. Feature noise, half-feature removal, and altered class balance produced 7.37%, 7.92%, and 8.47%, respectively. Replacing the primary predictor with a decision tree produced 15.77%. Missing 20% of observations produced 6.16%, while reducing cadence to 2 h produced 5.64%. Transfer

Table 3. Frozen robustness and negative-result summary.

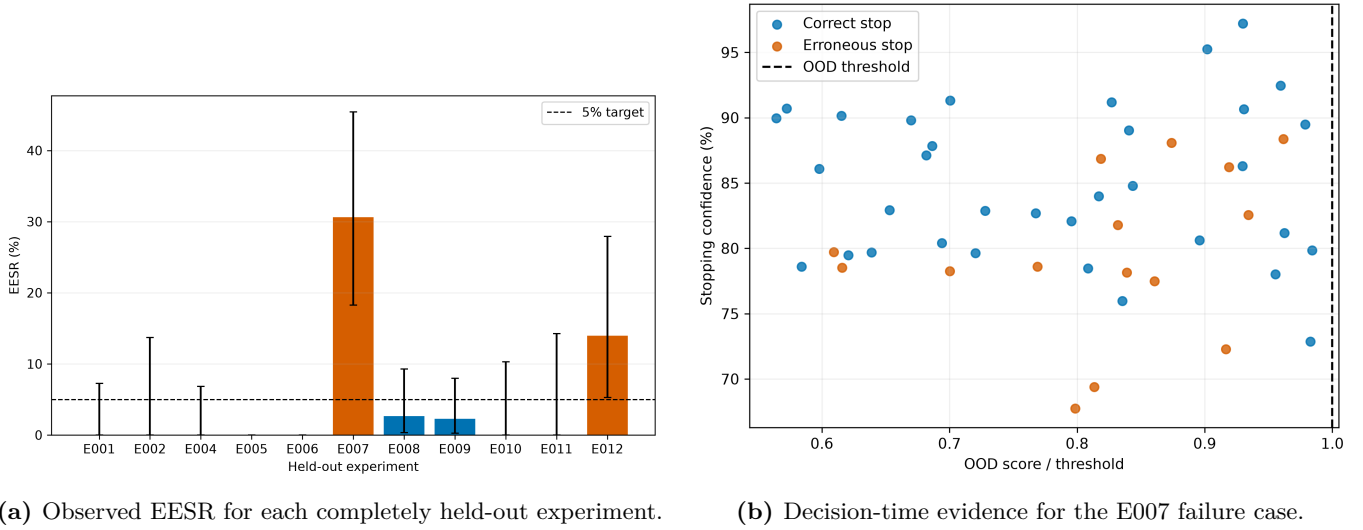
Condition	EESR	Savings
Primary RPE policy	5.69%	12.26%
Irregular sampling	10.61%	—
0.25-SD feature noise	7.37%	—
Half-feature removal	7.92%	—
Altered class balance	8.47%	—
Decision-tree predictor	15.77%	—
20% missing observations	6.16%	—
2 h cadence	5.64%	—
Lens endpoint transfer	17.82%	6.43%

to the Lens endpoint failed at 17.82% EESR with only 6.43% savings. These results are retained as evidence rather than treated as tuning opportunities.

12 Discussion

12.1 Early prediction is not safe termination

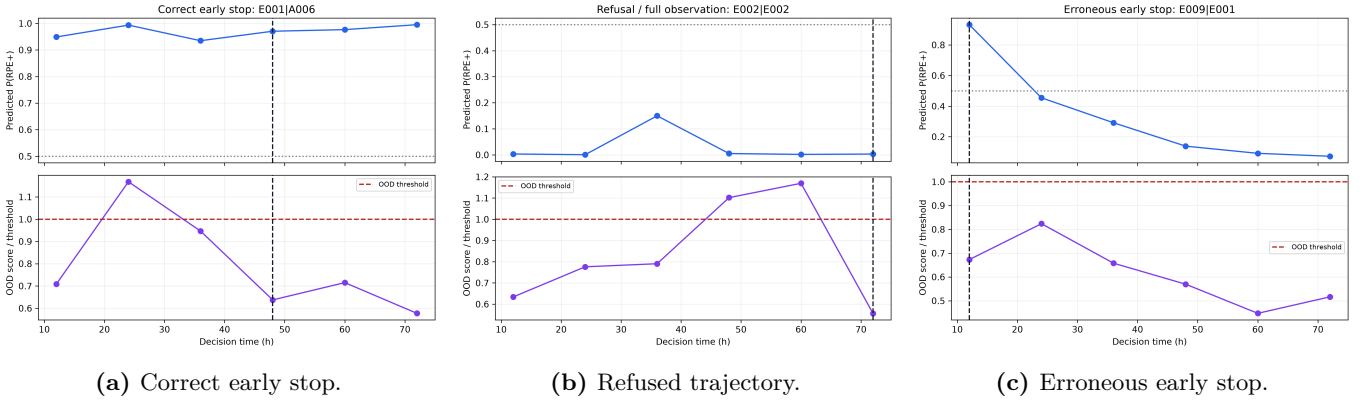
The source study established that longitudinal morphology contains early endpoint information. The present results show that converting such informa-



(a) Observed EESR for each completely held-out experiment.

(b) Decision-time evidence for the E007 failure case.

Figure 4. Experiment-level heterogeneity under the frozen 5% RPE policy. Aggregate EESR was 5.69%, but E007 reached 30.61% among early stops and was not rejected by the frozen trajectory-level familiarity rule. This is a central limitation, not an excluded outlier.



(a) Correct early stop.

(b) Refused trajectory.

(c) Erroneous early stop.

Figure 5. Representative frozen held-out trajectories. The examples expose the intended decision contract: early automation when calibrated evidence passes, continued observation after refusal, and an explicit failure that remains in the audit record.

tion into an intervention requires a distinct evaluation. Full GATES had lower observed EESR than the tested alternatives at its selected operating point, but this improvement came with selective coverage. Its value lies in the ability to continue or abstain when evidence does not pass, not in forcing every trajectory into an early decision.

12.2 What E007 teaches

E007 demonstrates that acceptable aggregate performance can coexist with severe experiment-specific failure. Whole-experiment separation reveals this failure but does not solve it. The current feature-distance rule recognized some unfamiliar trajectories, yet E007's errors were confidently inside its accepted region. A familiarity metric based on marginal feature distance can miss conditional or calibration shift. Larger calibration populations and explicit group-conditional validation are needed

before intervention.

12.3 Interpreting coverage and savings

Coverage describes the fraction of units for which early automation is permitted at a particular operating point. Lower coverage may be appropriate when the cost of a false stop is high. Likewise, 12.26% savings means that, in retrospective replay, scheduled observations after frozen decision times would not have been required for early-stopped units. It does not establish a 12.26% reduction in experiment duration, instrument occupancy, labor, storage cost, or money. Those consequences require prospective operational measurement.

12.4 Exploratory experiment-level gate

An optional extension summarized each experiment using information available by 12 h. At the 5% operating point, it refused E007 and E008, prevented 17

errors, reduced EESR to 2.54%, and retained 7.88% savings. At 10%, however, it accepted all seven unsafe experiments, refused all four safe experiments, prevented no errors, and worsened EESR to 16.34%. This instability precludes including the gate in the principal method claim.

13 Limitations and future work

The evidence is retrospective and comes from one published medaka dataset with only 11 independent experiments. There is no human-organoid, independent-laboratory, prospective, intervention, deployment, or direct organ-on-chip validation. The exact unit-level interval does not capture all between-experiment biological uncertainty. The refusal rule can miss conditional and calibration failures. Coverage and refusal are class-asymmetric. The policy provides neither anytime-valid nor distribution-free nor group-conditional control. No operational study has measured labor, microscope, storage, duration, or cost savings.

The immediate next experiment is a preregistered prospective evaluation in independent laboratories that preserves complete experiment boundaries and logs every candidate decision without acting on it. Subsequent work should evaluate human organoids and direct organ-on-chip trajectories, enlarge calibration populations, examine class-conditional reliability, strengthen missingness and cadence robustness, and develop formal anytime-valid group-conditional control before any stopping intervention.

14 Reproducibility

The public workflow pins the Zenodo record, filename, 400,059,241-byte file size, MD5, and SHA-256. Protocols, seeds, and the software environment are frozen. Machine-readable artifacts include per-unit terminal decisions, per-time predictions, exact and experiment-bootstrap intervals, operating frontiers, robustness analyses, figure-source mappings, and release manifests. A clean clone reproduces the evidence with the locked environment and a single task-runner command. The interactive demonstration reads frozen evidence and does not retrain. The complete public release is available at github.com/Atharva12081/GATES.

The v1 evidence audit contained an audit-only yes/no decoding defect in descriptive prevalence metadata. V2 corrected that metadata. The headline tables were independently recomputed and remained numerically unchanged; the method, frozen decisions, and principal result did not change.

15 Conclusion

Early prediction is not the same as safe early termination. GATES shows how longitudinal biological prediction can be recast as an experiment-disjoint selective decision that may stop, continue, or refuse. On frozen held-out medaka organoid trajectories, refusal reduced observed premature error while retaining measurable estimated savings, but experiment-level failures and wide uncertainty remain. The contribution is therefore an auditable decision framework and a transparent retrospective evaluation—not a guarantee of safe deployment.

References

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A Frozen protocol and decision semantics

The frozen analysis evaluates the complete sequential policy, not isolated per-time classifiers. For every held-out experiment, preprocessing statistics, predictor parameters, confidence thresholds, and familiarity thresholds are estimated without access to any of its wells, frames, or labels. Each well contributes one terminal record to aggregate evaluation. The same trajectory cannot contribute to more than one train, calibration, or test role within a rotation.

Table 4. Operational interpretation of the three terminal actions in the retrospective policy.

Action	Frozen rule	Evaluation consequence
STOP	Confidence passes and the prefix remains inside the fitted familiarity boundary.	The endpoint prediction is committed before 72 h; an incorrect label contributes one erroneous early stop.
CONTINUE	Available evidence does not pass the stopping rule.	The next scheduled observation is consumed and the policy is evaluated again at a later candidate time.
ABSTAIN	The trajectory crosses the frozen feature-distance refusal threshold.	Early automation is disabled for that unit and observation continues through the endpoint.

Table 5. Protocol invariants applied to every experiment-held-out rotation.

Invariant	Implementation consequence
Complete-experiment separation	No well, frame, prefix, or endpoint from the test experiment is used for preprocessing, fitting, calibration, or refusal-threshold selection.
Prefix-only information	Features at candidate time t summarize only measurements available by t ; future observations and endpoint labels are excluded.
Identifier and availability exclusion	Experiment identifiers and observation-availability variables are not predictor inputs.
One held-out terminal record	Each of the 988 wells contributes exactly one final action and, when applicable, one early endpoint prediction.
Frozen empirical interpretation	The 5% label is a calibration target; evaluation reports the observed full-policy outcome rather than a certified guarantee.

B Per-experiment frozen results

Table 6 reports the complete held-out RPE results at the primary operating point. The distribution is strongly heterogeneous: E005 and E006 have no early stops, seven experiments have no erroneous early stop, and E007 dominates the aggregate error count. Reporting only the pooled 25/439 result would obscure this structure.

Table 6. Full GATES at the frozen 5% operating point for each completely held-out experiment.

Experiment	Units	Early stops	Errors	EESR	Coverage	Savings
E001	80	49	0	0.00%	61.25%	16.67%
E002	92	25	0	0.00%	27.17%	4.53%
E004	96	52	0	0.00%	54.17%	9.03%
E005	92	0	0	0.00%	0.00%	0.00%
E006	95	0	0	0.00%	0.00%	0.00%
E007	93	49	15	30.61%	52.69%	30.29%
E008	91	75	2	2.67%	82.42%	16.67%
E009	90	88	2	2.27%	97.78%	27.22%
E010	92	34	0	0.00%	36.96%	6.16%
E011	90	24	0	0.00%	26.67%	8.89%
E012	77	43	6	13.95%	55.84%	17.53%
All	988	439	25	5.69%	44.43%	12.26%

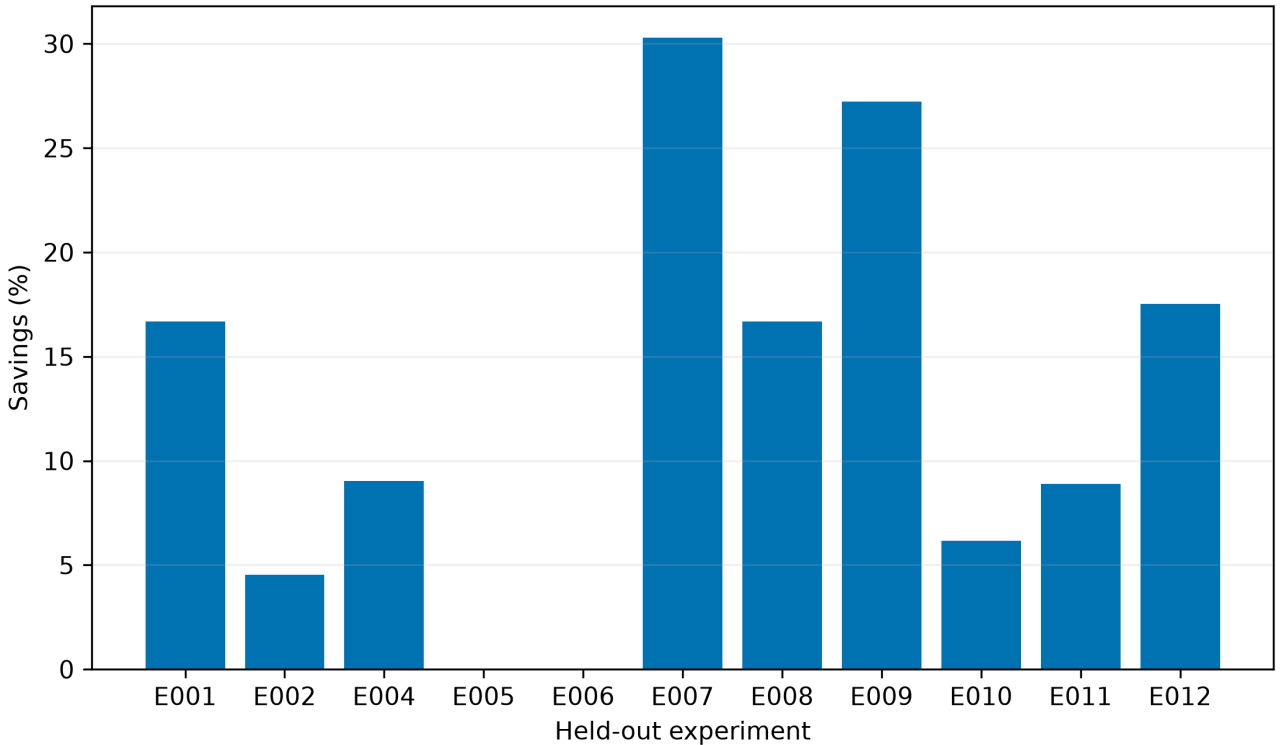


Figure 6. Retrospectively estimated observation savings for each held-out experiment under the frozen 5% policy. Savings and error vary together imperfectly: high estimated savings in E007 coexist with severe premature error, reinforcing the need to report the complete tradeoff.

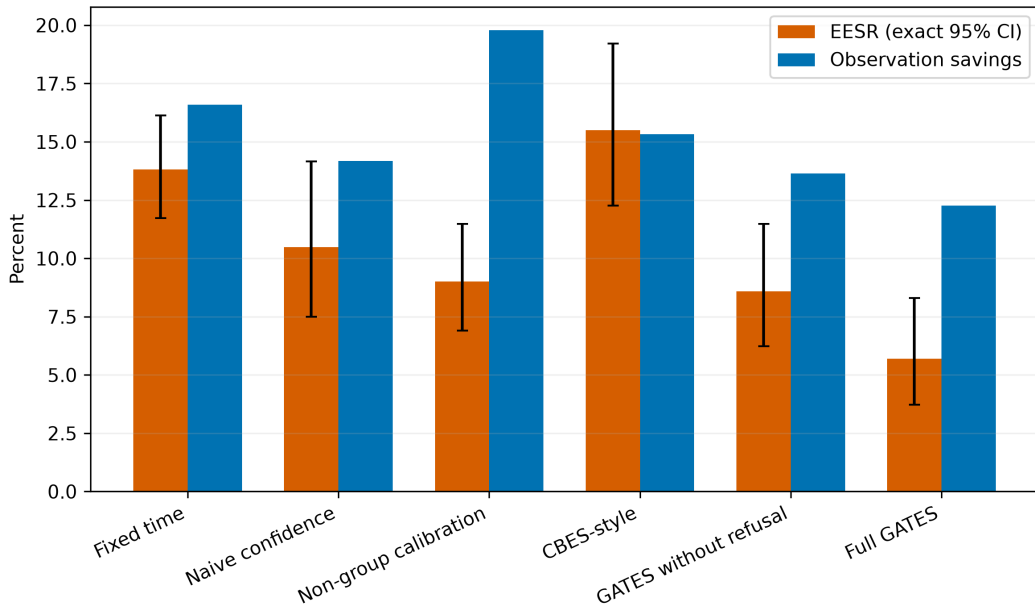
C Class-specific frozen-decision audit

The class audit in table 7 is computed only from frozen unit decisions; no model, threshold, split, or operating point was changed. Positive organoids were selected for early stopping less often and refused more often. This asymmetry is a limitation and motivates larger class-conditional calibration studies; it does not justify post hoc threshold adjustment.

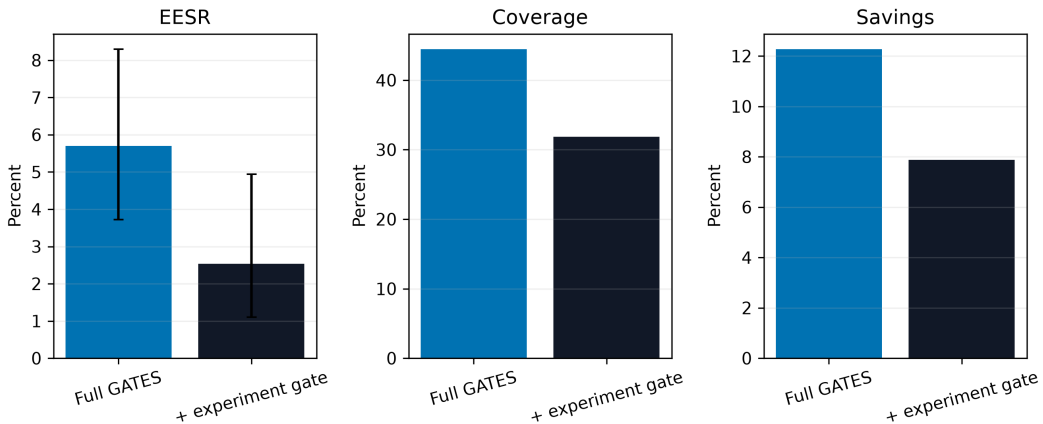
Table 7. Class-specific audit of frozen Full GATES decisions at the 5% operating point.

Endpoint class	Units	Early stops	Coverage	Refusals	Refusal rate	Errors	EESR among stops
RPE-negative	583	295	50.60%	14	2.40%	15 FP	5.08%
RPE-positive	405	144	35.56%	24	5.93%	10 FN	6.94%

D Baseline and exploratory-gate details



(a) Method comparison at the 5% operating-point label.



(b) Exploratory experiment gate at 5%.

Figure 7. Supplementary comparisons from frozen evidence. The baseline panel complements the frontiers but does not equate methods with different coverage. The experimental gate helped at 5%, yet its instability at 10% prevents promotion to the principal method.

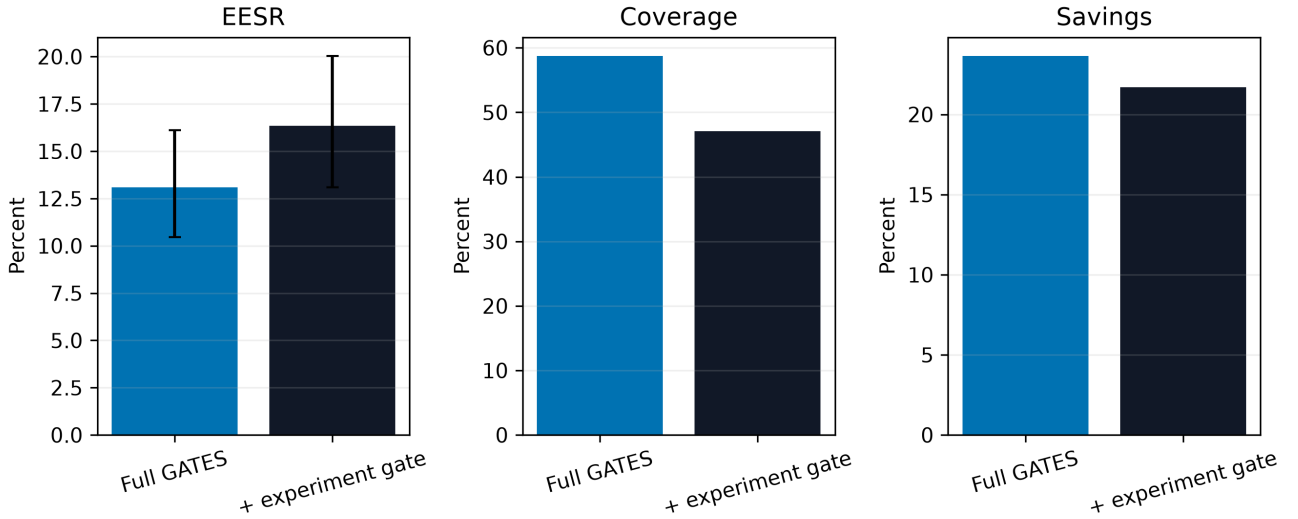


Figure 8. Failure of the exploratory experiment-level gate at the 10% operating point. It accepted all seven unsafe experiments, refused all four safe experiments, prevented no errors, and produced 16.34% EESR.

Table 8. Exploratory experiment-level gate. This extension is not part of Full GATES.

Metric	5% target	10% target
Early stops	315	465
Errors	8	76
EESR	2.54%	16.34%
Coverage	31.88%	47.06%
Savings	7.88%	21.69%

E Uncertainty record

Table 9. Frozen headline estimates and uncertainty intervals. The exact interval conditions on unit-level early stops; the experiment bootstrap resamples the 11 held-out experiments.

Metric	Point estimate	Interval
EESR	5.69% (25/439)	3.72–8.29% exact; 0.68–12.60% experiment bootstrap
Coverage	44.43% (439/988)	27.06–61.52% experiment bootstrap
Observation savings	12.26%	6.74–17.98% experiment bootstrap

The experiment-bootstrap intervals are the more relevant summary of between-experiment generalization, although they remain imprecise with only 11 groups. Neither interval supplies a prospective, anytime-valid, group-conditional, or distribution-free guarantee.

F Safety, reproducibility, and provenance

STOP means that the frozen retrospective research rule passed; it does not prescribe terminating a real experiment. CONTINUE means that the stopping threshold has not passed. ABSTAIN disables early automation for a trajectory outside the fitted feature-distance boundary and completes the 72 h protocol. Human oversight and prospective application-specific validation remain mandatory.

The headline selector is the RPE dataset, aggregate experiment, Full GATES, and 0.05 calibration target in the frozen master evidence table. Figure-to-table mappings, source-data checksums, decision records, and release hashes accompany the public repository. The retinal source data are licensed CC BY 4.0; project code is released under the MIT License.

Table 10. Reproducibility record included with the frozen release.

Item	Status
Pinned public source record and filename	Included
Byte size, MD5, and SHA-256 checks	Included
Locked dependency environment	Included
Frozen protocols and deterministic seeds	Included
Per-time predictions and per-unit decisions	Included
Figure-to-evidence mappings	Included
Release manifest and clean-clone replay	Included
Evidence-only interactive demonstration	Included

Requirements for a prospective study

The present evidence supports a shadow-mode next step rather than autonomous stopping. A preregistered evaluation should retain complete experiment boundaries, fit every calibration and refusal parameter before test decisions are revealed, and log actions at every candidate time without altering the experiment. Independent laboratories should be analyzed as distinct groups. The study should report class-specific coverage and refusal, cadence and missingness behavior, operational resource use, and every erroneous early stop. Human oversight remains necessary until application-specific prospective evidence and a suitable formal risk-control procedure are available. Human organoids and direct organ-on-chip trajectories require separate validation; the medaka results do not establish performance in either setting.

Table 11. Minimum evidence needed before considering an intervention study.

Requirement	Purpose
Preregistered frozen policy	Prevent threshold selection after viewing prospective outcomes.
Independent laboratories	Test transfer beyond the single released experimental source.
Shadow-mode sequential logging	Measure decisions prospectively without terminating acquisition.
Complete group separation	Preserve the biological generalization boundary.
Class-conditional reporting	Quantify asymmetric coverage, refusal, and error.
Operational measurements	Distinguish estimated observation savings from realized resource use.
Human oversight and stop rules	Protect experiments while evidence remains preliminary.
Direct target-system validation	Establish performance separately for human organoids or organ-on-chip trajectories.